Electrical Power and Energy Systems 157 (2024) 109863

Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes

Rule-based shields embedded safe reinforcement learning approach for electric vehicle charging control

Yuxiang Guan, Jin Zhang, Wenhao Ma, Liang Che *

College of Electrical and Information Engineering, Hunan University, Changsha, Hunan 410082, China

ARTICLE INFO

Keywords:
Reinforcement learning
Safe reinforcement learning
Electric vehicle
Distribution system
Vehicle-to-grid

ABSTRACT

The optimal and secure electric vehicles charging control can be formulated as a large-scale partially-observable constrained Markov decision process (PO-CMDP) problem with high levels of security risks and uncertainties. Such a problem is very difficult to be handled by optimization-theory-based methods. Data-driven methods, especially reinforcement learning (RL), are suitable for handling uncertainties but are poor in ensuring safety, which is unacceptable in power systems. Motivated by the fact that humans could be supervised by an expert when learning a new skill involving risk, this paper proposes a safe RL framework that embeds rule-based local and global shields in the loop of RL for supervising the actions of agents. The proposed framework not only strictly guarantees local and global security in the training and execution phases, but also helps the agent to find a more near-optimal policy. The effectiveness and efficiency are demonstrated by comparison with multiple baseline methods in the IEEE-33 node system.

1. Introduction

The International Energy Agency (IEA) estimates that the number of electric vehicles (EVs) will reach 125 million by 2030 [1]. The uncontrolled charging of large-scale EVs will impose large impacts on the grid, such as higher peak load, voltage variations and line overloads. Thus, optimal and efficient EV charging control is vital for ensuring the stable and secure operation of power systems. To this end, there are primarily two types of solutions: the optimization-based approach and the reinforcement learning (RL) approach.

The optimization-based approach typically uses classic optimization-theory-based algorithms [2,3] or heuristic algorithms [4]. However, the application of these methods encounters the following three key difficulties: 1) Difficult to handle uncertainties. The uncertainties arise from the EVs' charging behaviors including their arrival/departure times, charging starting/ending times, initial/target values of state of charge (SoC), the user's charging/discharging preferences; and the charging costs are also impacted by the time-of-use (TOU) prices. 2) Difficult to model. The application of optimization-based methods requires detailed and accurate models of the distribution grid, charging stations, and EVs' charging processes. 3) Difficult to solve. Such a nonlinear, nonconvex and large-scale problem is difficult to be resolved periodically in a short time and thus cannot meet the goal of real-time EV charging control.

Compared to the optimization approach, RL using a trained network is well suited to make fast online decisions for problems with the same distributional properties, and is able to deal with uncertainties and balance the immediate and long-term returns [5]. Early related works include [6] which applied Q-learning methods for the EV charging scheduling problem, and [7–9] which modeled a PO-MDP problem and solved it by using deep RL (DRL). Then, multi-agent RL is investigated. Refs. [10,11] proposed a multi-agent DRL method that optimized charging stations, PVs and energy storages. Ref. [12] proposed a centralized-training-decentralized-execution (CTDE) to manage EV charging. Ref. [13] proposed a fully decentralized MADRL-based control strategy to coordinate the charging process of EVs in a residential area. Moreover, when applying RL, researchers have begun to consider the complexities of power systems such as high dimensionality, mixed discrete–continuous actions, and privacy concerns. To handle the high dimensionality problem, advanced networks, e.g., graph neural networks [14,15], RNN [11], and LSTM [16], are used to extract environmental features and help agents to better understand the massive data. In the cases of mixed discrete–continuous action space, mathematical formulations of environments with discrete, continuous, and parametrized action spaces are established to solve EV charging scheduling problem [17]. Regarding privacy, Refs. [13] and [18] proposed selfish-collaborative multi-agent RL approach by only using EVs' local information and thus protect the EV users' privacy.

* Corresponding author.
E-mail addresses: guanyx@hnu.edu.cn (Y. Guan), zhangjin1994@hnu.edu.cn (J. Zhang), mawenhao@hnu.edu.cn (W. Ma), cheliang@hnu.edu.cn (L. Che).

https://doi.org/10.1016/j.ijepes.2024.109863
Received 11 October 2023; Received in revised form 2 January 2024; Accepted 8 February 2024
Available online 16 February 2024
0142-0615/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).

Nomenclature

Sets and indices

A	Action space of all agents
$c_i \in C$	The ith immediate constraint, and the set of constraints of all agents
i	Index of agents
$o_{i,t} \in O$	Local observation of agent i at time t, and the observation space
S	State space of all agents
t	Index of time instants

Parameters

$A^{\pi}/A^{\pi}_{c_i}$	The advantage functions of rewards and constraints of π
d_{π}	The discounted future state distribution of π
e_i^{max}/e_i^{min}	Upper/lower limits of charging/discharging power of agent i
N	Number of agents
Pr	State transfer probability
p^t	TOU price at time t
pv^t	Power of PV generation at time t
G_{km}/B_{km}	Conductivity/Susceptance of the branch from bus k to m
Res_{km}/X_{km}	Resistance/Reactance of the branch from bus k to m
SOC^i_{expect}	EV's SoC target at its departure time
SOC^i_{max}/SOC^i_{min}	Upper/lower SoC limits of EV i
t^i_{start}/t^i_{end}	Arrival and departure times of agent i
ΔU_j^{expect}	Acceptable range of voltage fluctuation (p.u.) of bus j
μ	EV charging efficiency
γ	Discount factor

Variables

$a_{i,t}$	Agent i's action at time t
$a_{i,t}^{shield}$	Agent i's action adjusted by the shield at time t
$\pi/\tilde{\pi}$	Policy and the optimal policy of an agent
$P_{j,t}/Q_{j,t}$	Active/Reactive powers at bus j, time t
$U_{j,t}/\theta_{j,t}$	Voltage and power factor angle at bus j, time t
$r_{i,t}/r_{i,t}^{shield}$	Agent i's reward and reward adjusted by the shield
$SOC^{i,t}$	SoC of EV agent i at time t

Although having been demonstrated to be a suitable way to address the studied problem, RL faces a key challenge—safety. During the training, the agent exploring the policy space will inevitably violate the safety criteria and be penalized many times. And even if the training converges, the agent is also likely to violate safety criteria in the execution. It is very difficult to guarantee the agents' actions strictly satisfy the relevant constraints. Obviously, this is a critical issue in power systems due to high security requirements.

To address this issue, safe RL has attracted increasing attention recently. It primarily includes three types. The first type is the Lagrangian-based safe RL approach, also known as the primal–dual policy optimization, which incorporates security constraints into the reward function [19–21]. This is the simplest but the least effective solution as it cannot strictly guarantee the policy's safety and leads to a large variance in the estimation of the expected cumulative reward, seriously affecting the agent's convergence [22]. The second type is the trust region theory-based safe RL method (e.g., constrained policy optimization (CPO) [23], projection-based CPO [24]). This method replaces the security constraint with a surrogate function [25,26], and transforms the non-convex problem into a solvable convex problem. However, it is computationally demanding and time-consuming, and only allows a very limited number of constraints. The third type resorts to external knowledge or human intervention to provide a safety guidance to the agent, which is a promising way of realizing safe RL [27–29]. But it is difficult and costly to obtain adequate and suitable knowledge or involve human experts when dealing with complex problems in power systems.

To address the studied EV charging problem and overcome the limitations in the previous safe RL discussed above, this paper proposes a safe RL approach. The proposed approach embedded a rule-based security protective scheme, called shield, in the RL's loop, for ensuring that the updated policy strictly satisfies the security requirement. In addition, the proposed framework is realized by a CTDE proximal policy optimization (PPO) algorithm to tackle the difficulty of partial observability among EV agents in the studied problem. The contribution is threefold:

(1) The optimal and efficient EV charging control is modeled as a partially-observable constrained Markov decision process (PO-CMDP) and solved by using RL. To overcome the critical drawback—security risk—in the application of RL in power systems, this paper proposes a safe RL framework that embeds rule-based shields in RL's loop, and thus strictly ensures security by using a physics-based mechanism and makes fast online decisions to the complex PO-CMDP problem by using the RL's data-driven feature. A comparison of this work with recent related studies is provided in Table 1.
(2) A CTDE architecture is established to handle the partial observability of the PO-CMDP. Compared with traditional centralized multi-agent algorithms with poor scalability and computational burden, and decentralized multi-agent algorithms with poor convergence under non-stationary environments [30], the proposed CTDE algorithm achieves a more optimal and secure policy by using global critics to coordinate the agents' policies under the supervision of shields.
(3) Both theoretical analysis and comparative experiments are performed. Compared with previous safe RL and optimization methods, the proposed approach has the merits of being able to strictly satisfy security constraints, higher computational efficiency, and less variance in the estimation of expected cumulative reward.

The remainder of the paper is organized as follows: Section 2 formulates the studied PO-CMDP problem, Section 3 proposes the safe RL approach with rule-based shields, Section 4 performs the numeric simulation and finally, Section V concludes the paper.

2. PO-CMDP formulation of the studied problem

Fig. 1 shows a distribution system with charging stations and PV generations. The studied problem involving different participants can be regarded as a two-level optimization problem with the EV and system levels. At the EV level, the EV users' objective is to minimize their charging costs while ensuring the charging meets their SoC targets. At the system level, the distribution system aims to dispatch all connected EVs for peak shaving while ensuring the operational security of the distribution grid. The decision variables include the EVs' charging and discharging powers. The randomness is derived from the EVs' behaviors including their arrival/departure times and initial SoCs.

The studied distribution system includes EV charging stations, PV generations and loads. Other distributed resources (e.g., energy storage and controllable loads) are not considered. This will not affect the effectiveness of the proposed method. The reason is given as follows. First, this paper focuses on safe and efficient EV charging control under

Table 1
Comparison of this work with recent related studies.

Remark	[7]	[8]	[9]	[10]	[12]	[13]	[18]	[19]	[25]	This work
Considering distribution grid security		✔	✔	✔						✔
No violation to security constraints of EVs and the grid										✔
Incorporating prior knowledge or physical rules in RL	✔			✔			✔			✔
Protecting EVs' privacy			✔		✔	✔				✔
Comparison with SOTA methods					✔			✔	✔	✔
Providing theoretical proof										✔

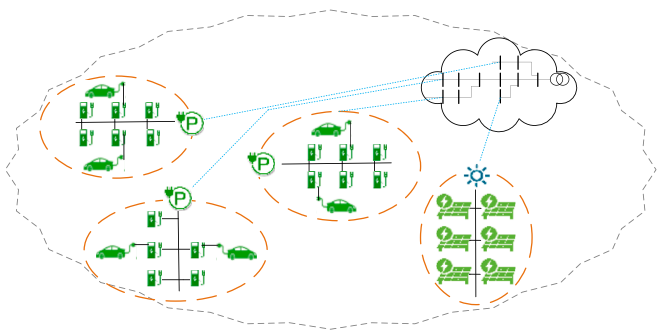

Fig. 1. A distribution system connecting EV and PV generations.

high level of PV penetration in distribution grids. Second, if other distributed resources (e.g., energy storage and controllable loads) are considered, the scheduling difficulty will decrease as the flexibility of these resources can help to achieve safe EV charging control [31,32]. While at the same time, the number of controlled agents and the dimensionality of states will generally increase, which can be effectively tackled by a centralized training and distributed execution algorithm architecture with good scalability.

In general, such a problem can be modeled as a CMDP which maximizes the following reward (1.1) while satisfying the immediate security constraints (1.2):

$$\max : J(\pi) = E_{\tau\ \pi}\left[\sum_{t=0}^{\infty}\gamma^t r_t\right] \tag{1.1}$$

$s.t.$

$$c_j(s_t, a_t) \le d_j, \qquad \forall j \tag{1.2}$$

where $J(\pi)$ is the expected discounted return of policy π, r_t is the agent's reward at time t, $\tau = (s_0, a_0, s_1, \cdots)$ denotes a trajectory, $E_{\tau\ \pi}[\bullet]$ is the expected value of the distribution over the trajectory τ, and $\tau\ \pi$ is short for $s_t\ d_\pi$, $a_t\ \pi(\bullet|s_t)$. The solution gives the following optimal policy:

$$\widetilde{\pi} = \underset{\pi\in\Pi_c}{\operatorname{argmax}} J(\pi) \tag{1.3}$$

Traditional approaches often treat an EV aggregation as an agent. But this will bring several problems: 1) difficulty in decomposing scheduling commands considering the uncertainty of EV behaviors [33]; 2) challenging the privacy of EVs [18]; and 3) inability to model complex interactions and competition among EVs. So, in order to address these challenges, the proposed approach models each EV as an agent that only uses its local information and interacts with other EV agents in all charging stations [10]. Thus, the local information of EVs, which is partially observable, is not shared among charging stations. So, the studied problem is modeled as the following PO-CMDP.

$$POCMDP = (S, A, O, Pr, \gamma, R, C) \tag{2}$$

The PO-CMDP consists of the six tuples: state space, action space, observation space, state transition probability, discount factor, reward function, and cost of constraint violation. The state space contains all the information characterizing the charging and discharging behavior of all EV agents. The observation space includes the information of a single EV. The action space encompasses all decision variables. The reward function is built based on the objective function of the optimization problem, and the cost of constraint violation represents the constraints of the optimization problem.

2.1. Local observation space and global state space

The local observation space O consists of $o_{i,t}$, which is the local observations of each EV agent i at time t:

$$o_{i,t} = \left[SOC^{i,t}, t^i_{start}, t^i_{end}, p^t, pv^t, t\right] \tag{3}$$

where $SOC^{i,t}$, t^i_{start} and t^i_{end} are agent i's local information, p^t and pv^t are the electricity price and PV power sent to the EV agents from an upper-level entity such as the distribution system operator. The global state s_t is formed by the union of the local information of all EV agents, i.e.,

$$s_t = o_{1,t} \cup o_{2,t} \cdots o_{N-1,t} \cup o_{N,t} \tag{4}$$

2.2. Action space

The action space A is formed by the upper/lower bounds of EVs' charging/discharging powers. Agent i's action space is:

$$a_{i,t} \in \left(e_i^{min}, e_i^{max}\right) \tag{5}$$

where positive and negative values mean the charging and discharging powers, respectively.

2.3. State transition probability

The state transition probability Pr includes the EV- and system-level probabilities. The EV-level state transition is represented by the EVs' charging dynamics modeled below, where μ is the charging efficiency.

$$SOC^{i,t+1} = SOC^{i,t} + \mu\frac{a_{i,t}}{W_i}\bullet\Delta t \qquad a_{i,t} \ge 0 \tag{6.1}$$

$$SOC^{i,t+1} = SOC^{i,t} + \frac{a_{i,t}}{\mu W_i}\bullet\Delta t \qquad a_{i,t} < 0 \tag{6.2}$$

where W_i is the EV battery capacity of agent i.

The distribution system-level state transition is denoted by the following Alternating Current power flow equation.

$$P_{k,t} = U_{k,t}\sum_{m=1}^{N_{bus}} U_{m,t}\left(G_{km}cos\theta_{km,t} + B_{km}sin\theta_{km,t}\right) \tag{7.1}$$

$$Q_{k,t} = U_{k,t}\sum_{m=1}^{N_{bus}} U_{m,t}\left(G_{km}sin\theta_{km,t} - B_{km}cos\theta_{km,t}\right) \tag{7.2}$$

where $\theta_{km,t}$ is the phase angle difference between bus k and bus m at time t and N_{bus} is the number of buses in the distribution network. In the studied problem, the EV- and system-level state transition probabilities are difficult to quantify precisely because it is difficult to obtain the large number of parameters in EVs and distribution network. So, the above

nonlinear state transition probability is unknown to EV agents. Hence, the proposed approach falls into the model-free RL, which means the control policy can only learn from the experience gained by interacting with the environment.

2.4. Reward

The reward R is the function of S and A. Since the studied problem involves different participants including the distribution system and EV users, R is composed of both the following system- and EV-level rewards:

(1) the system-level netload variation at time t:

$$r_t^{sys} = -\left| pv^t - \sum_{i=1}^{N} a_{i,t} \right| \tag{8.1}$$

(2) the EV agent i's charging cost under the TOU prices at time t:

$$r_{i,t}^{EV} = -p^t \bullet a_{i,t} \tag{8.2}$$

where r_t^{sys} represents the reward for coordinating all EV agents to absorb renewable energy at the system-level, and $r_{i,t}^{EV}$ represents the charging cost of the agent i at the EV level.

2.5. Cost of constraint violation

The cost of constraint violation (hereinafter denoted by C) [26] is also the function of S and A. Similar to the formulation of R above, C is composed of the following two aspects:

(1) the system-level c, i.e., the cost of violating the distribution network voltage constraints at time t:

$$c_t^{sys} = \frac{1}{N_{bus}} \sum_{j=1}^{N_{bus}} \left(U_{j,t} - U_j^{expect} \right)^2 \tag{9.1}$$

(2) the EV-level c, i.e., the cost of violating the EV agent i's security constraints at time t:

$$c_{i,t}^{EV} = \begin{cases} \left| SOC^{i,t} - SOC_{expect}^{i} \right| t \geq t_{end}^{i} \\ \left| SOC^{i,t} - clip\left(SOC^{i,t}, SOC_{min}^{i}, SOC_{max}^{i} \right) \right| else \end{cases} \tag{9.2}$$

where the two lines are respectively caused by the EV's failure to meet the SoC target (SOC_{expect}^{i}) at departure and by the EV's undercharging/overcharging during the charging process.

3. SRL approach embedded with rule-based shields

To solve the above PO-CMDP problem, this section first reviews two types of widely used safe RL methods and theoretically analyzes their weaknesses. Then, the proposed safe RL approach is proposed as a solution to the PO-CMDP. Its near optimality is proved. Finally, a PPO-based CTDE algorithm is developed to address the partial observability in PO-CMDP.

3.1. The limitations of existing safe RL approaches

3.1.1. Lagrangian-based safe RL method

The most common safe RL is the following Lagrangian-based method, also known as primal–dual policy optimization, which is currently the most applied method in the industry:

$$\left(\tilde{\pi}, \tilde{\xi} \right) = \underset{\xi \geq 0}{\arg\min} \max_{\pi_\theta \in \Pi_\theta} \left\{ J(\pi_\theta) - \xi J_c(\pi_\theta) \right\} \tag{10.1}$$

where $J_c(\pi_\theta)$ is the expected discounted return of the policy π_θ w.r.t. the cost of violating constraints, and $\xi J_c(\pi_\theta)$ is the Lagrangian term with lagrangian multiplier ξ. For brevity, the term in the pair of braces in (10.1), which is the new expected cumulative reward with constraints, is denoted by:

$$J_{Lag}(\pi_\theta) = J(\pi_\theta) - \xi J_c(\pi_\theta) \tag{10.2}$$

Although this method has been applied in power systems, it cannot guarantee the policy's safety. The primary reason is that it has the drawback of having a large variance in estimating the new expected cumulative reward, i.e., $\mathrm{Var}[J_{Lag}(\pi_\theta)]$ is a very large value. This will negatively impact the agent's convergence and lead to policies with both low rewards and poor satisfaction of constraints [22]. This is quantitively analyzed as follows.

For the convenience of discussion, let's consider the following case. The reward is$r_t \in$ [-1,1] for all trajectories, the cost of violating security constraints is$c_t = c_{Lag}$for trajectories violating constraints and $c_t = 0$ for trajectories without violating any constraint at time t, $\mathscr{P}_{Lag}$ is the probability of an agent violating constraints.

Lemma 1. *For the case given above, one has $c_t = c_{Lag}$ with the probability of $\mathscr{P}_{Lag}$, and $c_t = 0$ with the probability of $(1 - \mathscr{P}_{Lag})$. Thus, in order to avoid serious security issues, one must ensure that $\xi c_{Lag} \gg 1/\mathscr{P}_{Lag}$.*

Proof: It is a well-known fact that for any policy π_θ, $J(\pi_\theta) = E_{\tau\ \pi_\theta}[\sum_{t=0}^{\infty} \gamma^t r_t]$ can be reformulated as $J(\pi_\theta) = \frac{1}{1-\gamma} E_{\tau\ \pi_\theta}[r_t]$ [34]. So, $J_{Lag}(\pi_\theta)$ can be reformulated as:

$$J_{Lag}(\pi_\theta) = J(\pi_\theta) - \frac{1}{1-\gamma} \xi c_{Lag} \mathscr{P}_{Lag} \tag{11.1}$$

where $1/(1-\gamma)$ called the "effective time horizon" represents the number of steps the agent can foresee and is a fixed constant. The impact of the security constraint is denoted by the term $-\frac{1}{1-\gamma} \xi c_{Lag} \mathscr{P}_{Lag}$. If this term is ignored, i.e., $\xi c_{Lag} \ll 1/\mathscr{P}_{Lag}$, the agent will only focus on gaining the charging station's revenues, and this will result in security issues such as over-voltage and failing to meet SoC targets. Therefore, in order to avoid these security issues, one must ensure $\xi c_{Lag} \gg 1/\mathscr{P}_{Lag}$.

Theorem 1. *For the case given above, the estimation of the expected cumulative reward (10.2) will have a large variance, i.e.,* $\mathrm{Var}[J_{Lag}(\pi_\theta)]$ *is a very large value.*

Proof: For the value of $\mathrm{Var}[J_{Lag}(\pi_\theta)]$, according to [22], the following inequality holds:

$$\mathrm{Var}\left[J_{Lag}(\pi_\theta) \right] \geq \frac{1}{(1-\gamma)^2} \left(\left(\mathscr{P}_{Lag} \xi c_{Lag} \right)^2 \left(1 - \mathscr{P}_{Lag} \right) + \left(\xi c_{Lag} - \mathscr{P}_{Lag} \xi c_{Lag} \right)^2 \mathscr{P}_{Lag} \right)$$

$$= \frac{1}{(1-\gamma)^2} \mathscr{P}_{Lag} \left(\xi c_{Lag} \right)^2 - \frac{1}{(1-\gamma)^2} \left(\mathscr{P}_{Lag} \xi c_{Lag} \right)^2 \tag{11.2}$$

Firstly, Lemma 1 has proved that one should have $\xi c_{Lag} \gg 1/\mathscr{P}_{Lag}$ for avoiding serious security problems. Secondly, due to the large policy space caused by the multi-level (system-station-EVs) structure in the studied PO-CMDP problem, the probability $\mathscr{P}_{Lag}$ is generally very small, so ξc_{Lag} is a very large term. Due to these two facts, the term "$\frac{1}{(1-\gamma)^2} \mathscr{P}_{Lag} (\xi c_{Lag})^2$" has an exceptionally large value. As this term plays a dominant role on the right-hand side of (11.2), it will cause a very large value of $\mathrm{Var}[J_{Lag}(\pi_\theta)]$ on the left-hand side of (11.2).

The above theories are further explained as follows. Lemma 1 shows that, in order for the Lagrangian-based safe RL to enable the agent to learn policies without violating security constraints, the product of the Lagrangian multiplier and the constraint-violation cost should be large.

However, Theorem 1 proves that, an excessively large product will also have negative effects, e.g., introducing large variance in the estimation of expected cumulative reward and impacting convergence of the agents' training [22].

In sum, the Lagrangian-based method is the simplest safe RL but is unsuitable for problems with critical safety concerns.

3.1.2. Trust region theory-based safe RL method

Compared with the Lagrangian method, the trust region theory-based safe RL method formulated below replaces the security constraint with a surrogate function that evaluates $J_{c_i}(\pi_k)$ according to the samples collected from the current policy π_k [23].

$$\pi_{k+1} = \underset{\alpha\ \pi}{\operatorname{argmax}} E_{s\ d_{\pi_k}} [\widehat{A}^{\pi_k}(s,a)] \tag{12.1}$$

$s.t.$

$$J_{c_i}(\pi_k) + \frac{1}{1-\gamma} \underset{\alpha\ \pi}{E_{s\ d_{\pi_k}}} \left[\widehat{A}^{\pi_k}_{c_i}(s,a)\right] \le d_i \quad \forall i \tag{12.2}$$

$$E_{s\ d_{\pi_k}}[D_{KL}(\pi\|\pi_k)[s]] \le \epsilon^{\pi}_{\pi_k} \tag{12.3}$$

where $D_{KL}(\pi\|\pi_k)[s]$ is the total KL-divergence between policies π and π_k under state s and $\epsilon^{\pi}_{\pi_k}$ is the upper bound of KL distance between π and π_k. For the non-convex optimization problem in (12), CPO performs a first-order Taylor expansion of the objective function and a second-order estimation of the constraints, thereby transforming the non-convex problem into a solvable convex problem. However, this method is computationally intensive and time-consuming as it needs to use conjugate gradients for approximating the fisher information matrix when optimizing the new lower bound surrogate optimization function [35].

Therefore, the trust region theory-based safe RL method is difficult to be applied to the studied problem in this paper.

3.2. The proposed approach

3.2.1. Overall framework of the proposed approach

In general, when learning a new skill involving risk, humans tend to learn from an expert, who as a guardian intervenes in the learning and demonstrates correct actions when noticing that the learner encounters a danger [36]. Motivated by this, this paper proposes a safe RL framework embedded with rule-based shields depicted in Fig. 2, where the solid line represents the execution phase, and the solid line together with the dotted line represent the training phase.

The process of implementing the proposed approach is as follows:

Step 1: Based on the physical rules of the large-scale EV scheduling scenario, the local shield and the global shield are formed respectively, which includes three functions: action risk assessment, action adjustment and safety penalty.

Step 2: The actor of each EV agent uses local observation $o_{i,t}$ from the EV dispatching environment to make action $a_{i,t}$, and the critic of each EV agent receives local observations $o_{i,t}$ from all actor networks and forms the global state s_t.

Step 3: Rule-based shields adjusted the initial action $a_{i,t}$ to secure action $a^{shield}_{i,t}$ and applied it to the environment.

Step 4: Based on s_t, the critic calculates the advantage functions $\widehat{A}(s_t, a_{i,t})$ by the generalized advantage estimator (GAE) method [37] and returns it to the corresponding actor for updating, and critic itself updates based on the reward $r_{i,t}$ from the environment.

(a) $a_{i,t}$ is the agent's initial action mapped from the state space to the action space. So, $a_{i,t}$ does not need to consider security constraints. $a_{i,t}$ $\widetilde{\pi}$ can be learned from the trajectories obtained by interacting with the environment to maximize the cumulative reward as given in (13). Hence, in the proposed safe RL framework, $a_{i,t}$ only needs to maximize EV agents' revenues.

$$\widetilde{\pi} = \underset{\tau\ \pi}{\operatorname{argmax}} E_{\tau\ \pi}\left[\sum_{t=0}^{\infty} \gamma^t r_t\right] \tag{13}$$

(b) For the security constraints existing in PO-CMDP, the proposed rule-based shield corrects the initial action $a_{i,t}$ that only focuses on maximizing rewards to be a secure action $a^{shield}_{i,t}$ that strictly satisfies the security constraints given below.

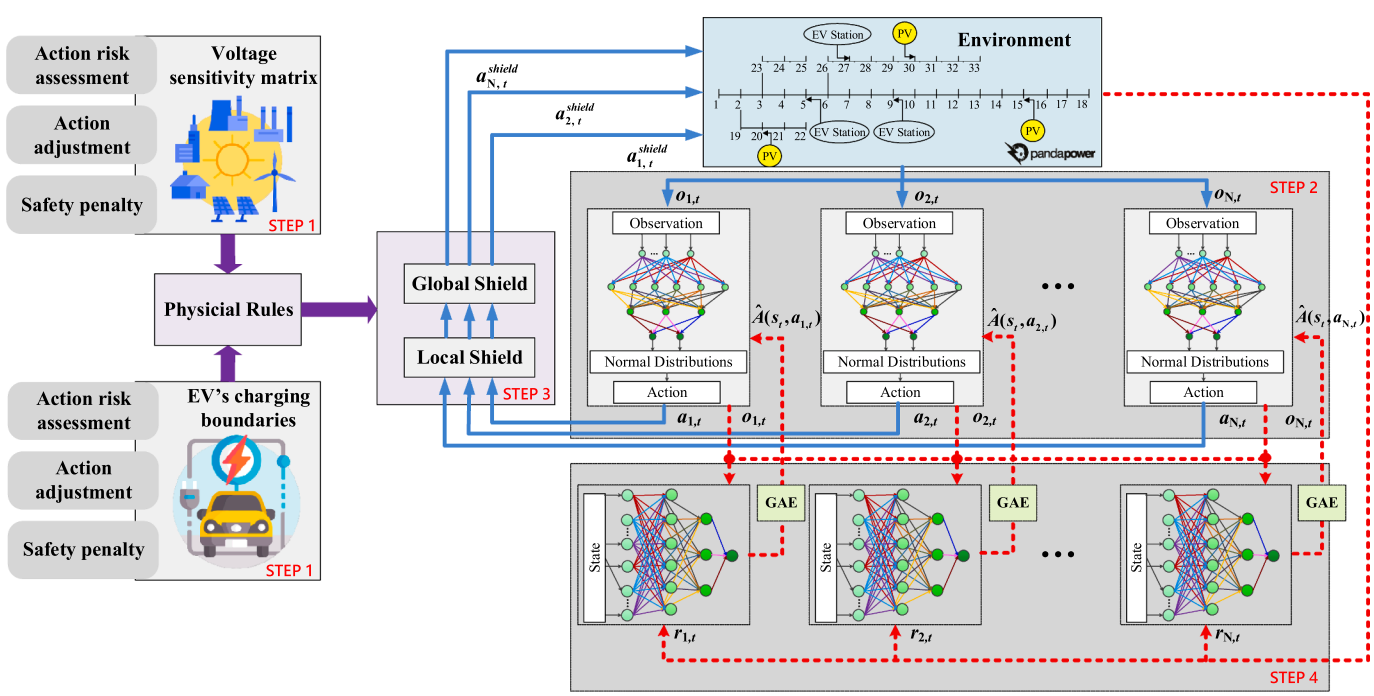

Fig. 2. The overall framework of the proposed approach.

$$a_{i,t}^{shield} = \text{argmin}\frac{1}{2}\|a - a_{i,t}\|^2 \tag{14.1}$$

$s.t.$

$$c_j(s_t, a_{i,t}) \le d_j, \qquad \forall j \tag{14.2}$$

The above formulation is explained as follows. One can firstly use the physical rules to establish the secure domain, which essentially creates a policy space ensuring strict satisfaction of security constraints at both EV level and system level. Then, within the secure domain, the proposed shield finds a new and secure action $a_{i,t}^{shield}$ that is close to the initial action $a_{i,t}$.

Regarding the proposed rule-based shields embedded safe RL approach, it is worth clarifying the following three points:

(a) The proposed approach is different from the "safety layer" methods mainly applied in the computer science discipline [36], which only applies to small-scale linear systems. The proposed approach establishes the action mapping from the rule-based local and global security domain, which helps the agents to explore in the early stage of training and maintain their continuity and stability of the agent's policy in the later stage of training in large-scale nonlinear systems.

(b) To avoid the problem of poor scalability and computational burden under traditional centralized multi-agent algorithms and poor convergence under non-stationary environments in decentralized multi-agent algorithms [30], the proposed approach will be implemented by a multi-agent PPO (MAPPO)-based CTDE algorithm discussed at the end of this section.

(c) As mentioned earlier, the proposed approach establishes a multi-agent framework (also found in [10]) and models the problem as a PO-CMDP by considering the facts that the partially-observable EV local information is not shared and the EVs have complex interaction and competition.

3.2.2. Constructing the shield by physical rules

As the first step in the proposed approach, physical rules are used to establish the secure domain, which is essentially a policy space that can guarantee that security constraints (including EV-level and distribution system-level security constraints) are strictly satisfied. This secure domain provides a basis for building the shield. The shield enables the agent to explore in the secure domain and finally converges to a policy that is secure and near-optimal. The shield consists of the local and global shields as follows.

(a) Local Shield for the EV level. To ensure the EV-level security (e.g., avoiding under/overcharging and meeting the SoC target), the secure boundary is set based on the EV's SoC, which is depicted by the upper/lower boundaries e^+ and e^- in Fig. 3. In essence, e^+ represents the case where the EV charges using maximum charging power until reaching its SoC target or upper SoC limit; while e^- refers to the converse case where the EV discharges to the lower SoC limit and finally charges for meeting its SoC target. The region between them is the practical secure domain for the EV's charging.

The local shield obtains the information of the current $SOC^{i,t}$ and time t from $o_{i,t}$, and delineates the practical secure domain for the $a_{i,t}$ of agent i. For EV agent i' action a_i, its local secure domain is:

$$\mathbb{D}_{local,i}^{shield} = \left\{a_{i,t} | SOC^{i,t} \in \Omega_{soc}^{i,t}, for\ all\ t\right\} \tag{15}$$

where $\Omega_{soc}^{i,t}$ is the time-varying SoC secure domain of EV agent i. $\mathbb{D}_{local,i}^{shield}$ is the set of feasible charging/discharging powers ensuring $SOC^{i,t}$ of EV agent i in the $\Omega_{soc}^{i,t}$ above at all times.

(b) Global Shield for the System Level. Similarly, the global secure domain delineates the operational domain in which the charging/discharging behaviors of large-scale EVs connected to multiple charging stations can satisfy the security requirements of the distribution system.

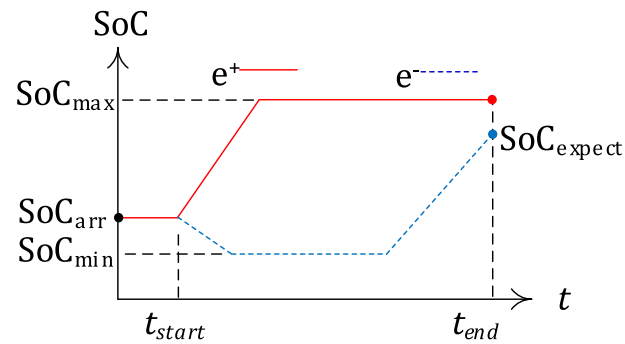

Fig. 3. An EV's charging and discharging boundaries.

Denote the real and reactive power fluctuations at bus m by $\Delta P_{m,t}$ and $\Delta Q_{m,t}$. Bus k will have the following voltage variation [38], where $T_{m,k,vol}^{P}$ and $T_{m,k,vol}^{Q}$ defined below are sensitivity factors between bus k's voltage and bus m's powers:

$$\Delta U_{k,t} = T_{m,k,vol}^{P} \bullet \Delta P_{m,t} + T_{m,k,vol}^{Q} \bullet \Delta Q_{m,t} \tag{16.1}$$

$$T_{m,k,vol}^{P} = \begin{cases} \sum_{j \in M_k^{set}} \frac{Res_{j-1j}}{U_0} & k \in M_m^{set} \\ \sum_{j \in M_m^{set}} \frac{Res_{j-1j}}{U_0} & m \in M_k^{set} \\ T_{m,F_k,vol}^{P} & else \end{cases} \tag{16.2}$$

$$T_{m,k,vol}^{Q} = \begin{cases} \sum_{j \in M_k^{set}} \frac{X_{j-1j}}{U_0} & k \in M_m^{set} \\ \sum_{j \in M_m^{set}} \frac{X_{j-1j}}{U_0} & m \in M_k^{set} \\ T_{m,F_k,vol}^{Q} & else \end{cases} \tag{16.3}$$

where M_m^{set} is the set of buses from bus m to the source bus of the distribution network, U_0 is the rated voltage of the source bus, and F_k is the first bus in the bus set M_k^{set} that connects other lateral branches.

For the charging station at bus m, its power is the net sum of the charging/discharging powers of all EVs in this station:

$$\Delta P_{m,t} = \sum_{i=1}^{N} a_{i,t} \tag{17}$$

For a bus k in the distribution network, its voltage fluctuation is caused by power fluctuations of all charging stations connected to the grid:

$$\Delta U_{k,t} = \sum_{m \in \Omega_{CS}} \left(T_{m,k,vol}^{P} + T_{m,k,vol}^{Q} tan\theta_m\right) \Delta P_{m,t} \tag{18}$$

where Ω_{CS} is the set of buses connecting charging stations. The charging stations' power fluctuations need to ensure $\Delta U_{k,t}$ in its acceptable range:

$$\Delta U_{k,t} \le \Delta U_k^{expect} \tag{19}$$

Hence, the global secure domain $\mathbb{D}_{global}^{shield}$ is constructed as:

$$\mathbb{D}_{global}^{shield} = \left\{\Delta P_{m,t} | \Delta U_{k,t} \le \Delta U_k^{expect},\ m \in \Omega_{CS}\right\} \tag{20}$$

where $\mathbb{D}_{global}^{shield}$ represents the set of charging station's power that ensures that the voltage variation $\Delta U_{k,t}$ at bus k is less than ΔU_k^{expect} at all times.

3.2.3. Shield-based safe RL approach

As discussed earlier, the rule-based shield embedded in the RL's loop

supervises the agent's policy as illustrated in Fig. 4. Take local shield as an example, when the local shield senses an unacceptable risk in the agent's action $a_{i,t}$, it tunes $a_{i,t}$ to be a secure action $a_{i,t}^{shield}$ and also adjusts the environment's reward $r_{i,t}$ to be $r_{i,t}^{shield}$ for imposing a safety penalty on the agent. The blue line represents the safe process, and the red line represents the risky process in Fig. 4, which includes the following three key steps, namely action risk assessment, action adjustment and safety penalty calculation.

(a) Action risk assessment. To measure the risk of actions, the following local and global safety indicators are constructed:

$$SI_{i,t}^{local} = \begin{cases} 0 a_{i,t} \in \mathbb{D}_{local,i}^{shield} \\ \dfrac{a_{i,t}}{clip\left(a_{i,t}, \mathbb{D}_{local,i}^{shield}\right)} else \end{cases} \tag{21.1}$$

$$SI_{t}^{global} = \begin{cases} 0 \Delta P_{m,t} \in \mathbb{D}_{global}^{shield} \\ \dfrac{\Delta P_{m,t}}{clip\left(\Delta P_{m,t}, \mathbb{D}_{global}^{shield}\right)} else \end{cases} \tag{21.2}$$

when $SI_{i,t}^{local}$ shows a security risk for EV i or SI_{t}^{global} warns a security risk in the distribution system, the action $a_{i,t}$ is considered to be risky and thus the shield will be triggered.

(b) Action adjustment. With the local and global rule-based secure domains constructed above, (14) can be reformulated as:

$$a_{i,t}^{shield} = \text{argmin} \frac{1}{2} \| a - a_{i,t} \|^2 \tag{22.1}$$

$$s.t.$$

$$a \in \mathbb{D}, \mathbb{D} = \mathbb{D}_{local,i}^{shield} \; or \; \mathbb{D}_{global}^{shield} \tag{22.2}$$

where $\mathbb{D}$ is the local and global secure domains. This reformulation simplifies the solving of the problem (14) and lays a foundation for the action adjustment in the following. As discussed earlier, the action mapping established from the rule-based local and global security domain can help the agents to explore in the early stage of training and maintain the continuity and stability of their policy in the later stage of training.

Before discussing the action adjustment proposed, we introduce two basic adjustment techniques. One is the random-search-based adjustment in (23), which randomly selects actions within the secure domain regardless of the initial policy. This technique is more conducive to exploring the entire secure domain, but fails to reflect the updated direction of the agent's current policy in (22.1).

$$a_{i,t}^{shield} = random(\mathbb{D}) \tag{23}$$

The other technique is the clip-based adjustment in (24). It ensures that the adjusted action satisfies the secure domain and is also close to the action given by the agent's exploration policy in (22.1), which helps to maintain the policy's continuity. However, without sufficient exploration of the secure domains, this can easily lead to sub-optimal policies.

$$a_{i,t}^{shield} = clip\left(a_{i,t}, \mathbb{D}\right) \tag{24}$$

Considering the above two limitations, this paper develops the following action adjustment technique to be applied in the proposed shield framework:

$$a_{i,t}^{shield} = clip\left[(1-\mathscr{T}) random(\mathbb{D}) + \mathscr{T} clip\left(a_{i,t}, \mathbb{D}\right), \mathbb{D}\right] \tag{25}$$

where $\mathscr{T}$ is the normalized number of training phase ($0 \leq \mathscr{T} < 1$). Under this action adjustment approach, the first term in the brackets ensures that in the early stage of training, the shield helps the agent to explore and find infrequent state-action pairs with lower entropy values by randomly selecting actions within the secure domain. And in the later stage of training, the shield focuses on the neighborhood of the current policy of the agent specified by the second term to maintain the continuity and stability of the agent's policy.

(c) Safety penalty. When the shield evaluates that $\boldsymbol{a_{i,t}}$ is risky based on the safety indicators $\boldsymbol{SI_{i,t}^{local}}$ or $\boldsymbol{SI_{t}^{global}}$, after the action adjustment, the shield computes the following penalty according to the degree of violation of the safety indicators of risky actions $\boldsymbol{a_{i,t}}$.

$$R^{shield} = -\mathbb{F}\left(SI_{i,t}^{local}, SI_{t}^{global}\right) \tag{26}$$

where $\mathbb{F}$ is a nonlinear mapping for calculating the modified penalty R^{shield} based on the safety indicators. The safety penalty and the reward in (8) are of the same order of magnitude. Hence, by utilizing the nonlinear mapping in [39], the agent will obtain the environment's original reward when $a_{i,t}$ is safe (i.e., in the secure domain), be slightly warned when $a_{i,t}$ is exceeding the domain's boundary, and be strongly penalized when $a_{i,t}$ is risky (i.e., far beyond the boundary) in Fig. 5.

3.2.4. Near-optimality

Compared with the earlier-reviewed two safe RL approaches, the proposed approach can achieve better performance. This is primarily

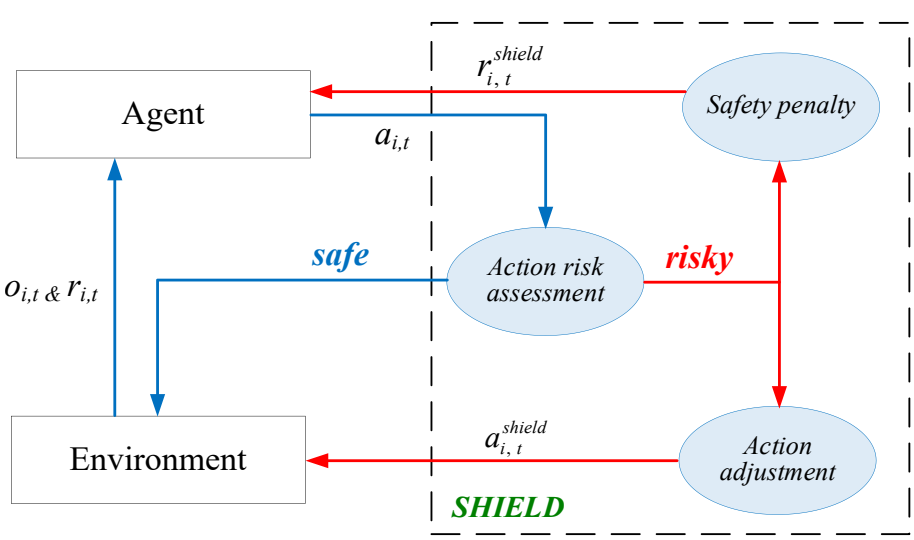

Fig. 4. Illustration of the shield's three functions.

due to the following two reasons.

(a) Compared with the Lagrangian-based method, the proposed method maximizes the cumulative reward without the need to directly handle the constraints. This avoids the Lagrangian method's excessive variance problem (discussed in Theorem 1) and therefore is more likely to train a more optimal policy than the Lagrangian method. This is proved in the following Theorem 2.

Theorem 2. *Given the case mentioned in Theorem 1, the variance of estimating the expected cumulative reward in the proposed method is much smaller than that in the Lagrangian method, i.e.,* $\mathrm{Var}[J_{shield}(\pi_\theta)] < \mathrm{Var}[J_{Lag}(\pi_\theta)]$.

Proof: Please refer to the Appendix.

(b) Compared with the trust region theory-based method, the proposed approach firstly uses the rule-based local and global security domains (15) and (18) to strictly guarantee the satisfaction of local and global constraints, and then uses the action adjustment technique (25) to convert risky actions to safe ones in both training and execution phases. This avoids the need to calculate conjugate gradients for approximating the fisher information matrix, and therefore is much more computationally efficient than the trust region theory-based method.

3.3. Multi-agent CTDE architecture with the shield

To deal with the PO-CMDP problem, the proposed shield-based safe RL approach is implemented by a multi-agent PPO (MAPPO)-based CTDE algorithm in this subsection.

The proposed MAPPO-based algorithm for the safe RL approach is given in Algorithm 1, in which the actor and critic networks have the following optimization objective [40]:

$$L(\theta) = \frac{1}{BN}\sum_{b=1}^{B}\sum_{i=1}^{N}\min\left\{\begin{array}{c}\frac{\pi_\theta(a_{i,t}|o_{i,t})}{\pi_\theta^{old}(a_{i,t}|o_{i,t})}\widehat{A}(s_t,a_{i,t}),\\ clip\left(\frac{\pi_\theta(a_{i,t}|o_{i,t})}{\pi_\theta^{old}(a_{i,t}|o_{i,t})},1-\varepsilon,1+\varepsilon\right)\widehat{A}(s_t,a_{i,t})\end{array}\right\}$$

$$+\sigma\frac{1}{BN}\sum_{b=1}^{B}\sum_{i=1}^{N}entropy\left(\pi_\theta\left(\bullet|o_{i,t}\right)\right) \tag{27.1}$$

$$L(\phi) = \frac{1}{BN}\sum_{b=1}^{B}\sum_{i=1}^{N}\max\left\{\begin{array}{c}(V_\phi-\widehat{r}_t)^2,\\ \left(clip\left(V_\phi,V_\phi^{old}-\varepsilon,V_\phi^{old}+\varepsilon\right)-\widehat{r}_t\right)^2\end{array}\right\} \tag{27.2}$$

where B is the batch size, *entropy* is the entropy of the policy, $\widehat{r}_t$ is the discounted reward-to-go, and σ is the hyperparameter adjusting the entropy coefficient.

The key steps in Algorithm 1 are further explained as follows:

Step 1: Based on the physical rules of the large-scale EV scheduling scenario, the local security domain and the global security domain are formed respectively.

Step 2: Initialize the environment and network parameters, and the algorithm's parameters including learning rate, maximum training times, batch size, and number of agents.

Step 3: All agents execute actions based on local observations.

Step 4: The local and global Shield evaluates the actions of agents in turn. If there is a risk, Shield will use secure actions to correct the risky actions.

Step 5: All actions are applied to the environment and the trajectories are stored in the experience buffer.

Step 6: The actor network and critic network are updated based on experience buffer.

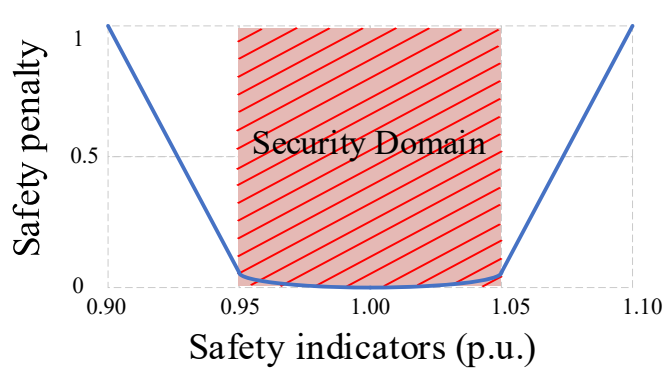

Fig. 5. Illustration of the nonlinear mapping of safety penalty.

4. Experiments

4.1. Experimental settings

The simulation is performed on a modified IEEE-33 bus distribution system shown in Fig. 6. PV generations are located at buses 15, 20, and 30. Three charging stations are connected at buses 5, 9, and 27 with 3, 3, and 4 EVs, respectively. In a station, each charger has the maximum charging/discharging powers of ± 80 kW, and the reactive powers are obtained based on 0.9 power factor. The following security criteria are considered: the maximum-allowed voltage deviation is 10 % in the distribution grid, each EV has the SoC limits [10 %, 95 %] and the SoC target 95 %. To simplify the state space to accelerate the training, this paper adopts the same assumptions as other studies [41], all EVs have the same battery capacity of 60 kWh. The load and PV output data are obtained from the California Independent System Operator (CAISO) [42]. The simulation time horizon is one day which is a complete EV charging cycle. The control time granularity is 30 min.

For the EV-level SoC constraint only, the indicator SOC_{mse} given below is used for fairly comparing the results among different methods.

$$SOC_{mse} = \frac{1}{N}\sum_{i=1}^{N}\left(SOC^{i,t} - SOC^{i}_{expect}\right)^2 \tag{28}$$

The EVs' charging behaviors are shown in Table 2. According to [43], the EVs' travelling data in this table is sampled from Gaussian distribution for incorporating the EVs' uncertainty in the simulation. It should be noted that the EVs' travelling data under other probability distributions can also be used here. TOU prices are given in Table 3.

In Algorithm 1, the structures of both Actor and Critic use fully-connected layers [26]. The network parameters are initialized orthogonally, updated for K = 6450 episodes, and evaluated every 5 episodes. Other hyperparameters are given in Table 4. 32 parallel environments are used to collect experience.

The simulations are performed in Python 3.6 and Pytorch 1.6 in a Dell 7920 T workstation with 6248R CPU, RTX3090 GPU, and 512 GB RAM.

4.2. Benchmark methods

For validation, the proposed approach (shield-embedded SRL) is benchmarked by comparing its performance with those of five state-of-the-art benchmark methods, which are listed as #1∼#5 in Table 5 and explained below. Comparison is conducted based on the reward terms (8) and the constraints (9) in both the training and execution phases.

(a) **Base**—The uncontrolled EV charging case mostly found in the industry, which charges an EV by following its maximum-allowed power curve until reaching its SoC target.

(b) **OPT**—An optimal charging method using evolutionary algorithm. Due to the partial observability in the PO-CMDP, this

Algorithm 1: Safe Multi-Agent Proximal Policy Optimization

Constructing local/global security domains $\mathbb{D}_{local,i}^{shield}/\mathbb{D}_{global}^{shield}$ based on physical rules (15) and (20).
Initialize the environment.
Initialize the policy network and value network parameters.
Set learning rate, maximum episodes K, batch size B, number of agents N.
for $k = 1, \dots, K$ **do**
  Clearing the experience buffer $\widehat{\mathcal{D}}$
  for $b = 1, \dots, B$ **do**
    while t is less than t_{end}^{i} **do**
      for $i = 1, \dots, N$ **do**
        $a_{i,t} = \arg\max_{a \in A} E_{\tau \sim \pi}[\sum_{t=0}^{\infty} \gamma^t r_t]$
        Local shield calculates $SI_{i,t}^{local}$ by (21.1)
        if $\boldsymbol{SI_{i,t}^{local}} \neq 0$ **then**
          $a_{i,t}^{shield} = clip\left[(1-\mathcal{T})random\left(\mathbb{D}_{local,i}^{shield}\right) + \mathcal{T}clip\left(a_{i,t}, \mathbb{D}_{local,i}^{shield}\right), \mathbb{D}_{local,i}^{shield}\right]$
          calculate penalty by (26)
        end if
      end for
      Global shield calculates SI_{t}^{global} by (21.2)
      if $\boldsymbol{SI_{t}^{global}} \neq 0$ **then**
        for $i = 1, \dots, N$ **do**
          $a_{i,t}^{shield} = clip\left[(1-\mathcal{T})random\left(\mathbb{D}_{global}^{shield}\right) + \mathcal{T}clip\left(a_{i,t}, \mathbb{D}_{global}^{shield}\right), \mathbb{D}_{global}^{shield}\right]$
        end for
        calculate penalty by (26)
      end if
      All agents execute their actions
      $o_{i,t} \leftarrow o_{i,t+1}$
    end while
    Store the trajectory τ in $\widehat{\mathcal{D}}$
    Calculate $\hat{A}(s_t, a_{i,t})$ by $\widehat{\mathcal{D}}$ using GAE method
  end for
  Update actor network by (27.1) and critic network by (27.2)
end for

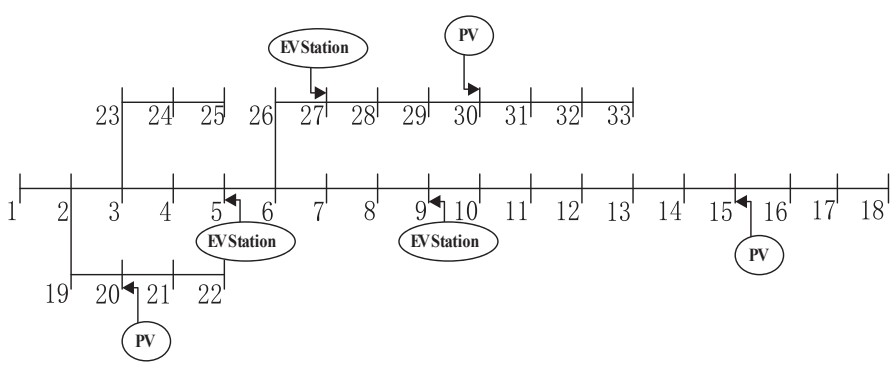

Fig. 6. Distribution network in the simulations.

Table 2
Probability distributions of EVs' travelling information.

Type	Probability distribution	Limits
Arrival time	$\mathcal{N}(8,1^2)$	[7,9]
Departure time	$\mathcal{N}(18,1^2)$	[17,19]
Initial SoC	$\mathcal{N}(0.2,(0.1)^2)$	[0.1, 0.3]

Table 3
TOU prices used in the simulation.

Type	Time period	TOU prices
Peak	10:00–15:00, 18:00–21:00	1.2710
Flat	7:00–10:00, 21:00–23:00	0.7523
Valley	23:00–7:00, 15:00–18:00	0.2849

Table 4
Hyperparameters used in the simulation.

Parameter	Value	Description
N	10	Number of agents
K	6450	Maximum episodes
C	32	Number of parallel environments
B	128	Batch size
$\alpha^{actor}, \alpha^{critic}$	0.0005	Learning rate
seed	1	Random seed
ppo_epoch	15	Update frequency
gae_lambda	0.98	Coefficient in GAE
σ	0.005	Entropy coefficient
ε	0.2	Clipping coefficient
μ	0.92	Charge/discharge efficiency
γ	0.99	Discount factor

method can only consider the local objective and security constraints in charging stations.

(c) **PRL**—MAPPO method that adds the constraint to the reward via the following penalty function.

$$r_t = r_t^{sys} + r_{i,t}^{EV} + \xi_U \bullet c_t^{sys} + \xi_{soc} \bullet c_{i,t}^{EV} \tag{29}$$

(d) **PPO-L**—Lagrangian-based SRL method with the PPO algorithm. Since the Lagrangian method is difficult to handle the partial observability in the PO-CMDP, this method can only consider the local objective and constraints in charging stations.

(e) **CPO**—Trust domain theory-based SRL method. Since CPO has difficulty in handling multi-level constraints, this method can only consider the local objective and security constraints in charging stations.

4.3. Experimentation in the training phase

The results in the training phase are compared in Figs. 7-10, where the horizontal axes represent the entire training process and each point denotes a single charging cycle (episode). The total reward is not compared as these methods do not have a uniform form of rewards (e.g., PRL and PPO-L adds security constraints to the reward function in different ways, and the proposed approach puts security constraints into the shield).

Fig. 7 compares the average charging cost (8.1) of the ten agents, which shows the results of "Base > PRL > OPT > PPO-L > CPO≈ the proposed approach". The reason is that, compared with Base, OPT reduces the cost by charging the EV under low TOU prices, PRL, PPO-L and CPO actively respond to the TOU price variations, and the proposed approach finds a more optimal charging policy with the protection of the local shield.

Fig. 7 compares the average EV charging costs (8.1). Among the six methods, only Base does not consider local objective in Table 5, which leads to the highest charging cost. Other methods reduce the cost by actively responding to the TOU prices. Under the proposed approach, the agent only needs to maximize local objectives with the protection of the local shield, and does not need to find a balance between objectives and constraints (which is needed under PRL and PPO-L). This allows the proposed method can find a more optimal charging policy. This is demonstrated by the following results shown in Fig. 7: "Base > PRL > OPT > PPO-L > CPO ≈ the proposed approach". The fluctuation of the curve is caused by the randomness in the EVs' behaviors.

Fig. 8 compares the SoC deviation (28) at the EVs' departure times. The OPT method determines analytical solutions, the Base method charges EVs to the target SoC at maximum power, and the proposed approach with local shield satisfies the local safety constraints. So, under these three methods, the indicators SOC_{mse} all converge to 0 in Fig. 8, which proved that the EV's SoC constraint is strictly satisfied. PPO-L, PRL and CPO decrease SOC_{mse}, but still fail to satisfy the EV's SoC constraint. The failure of PPO-L and PRL can be attributed to the significant variance problem arising from adding constraints to the reward function which poses challenges for agent convergence. On the other hand, CPO fails because it has limited scalability and is difficult to handle multi-level multi-security constraints. This can be clearly seen in Fig. 8 that during the interval of [18,20], SOC_{mse} of PPO-L, CPO and PRL still do not converge to 0.

Figs. 7-8 demonstrate the tradeoff between improving rewards and satisfying constraints. Compared with PRL, PPO-L and CPO in which the agent needs to constantly coordinate the local objective (optimizing cost) and the local security constraint (meeting SoC target), the proposed approach ensures the local security constraint by using the local shield so that the agent only needs to focus on exploring in the secure domain to optimize the cost. As a result, the proposed approach's results not only satisfy the security criteria but also achieve more optimal and stable cost than PPO-L and CPO as seen in Figs. 7-8.

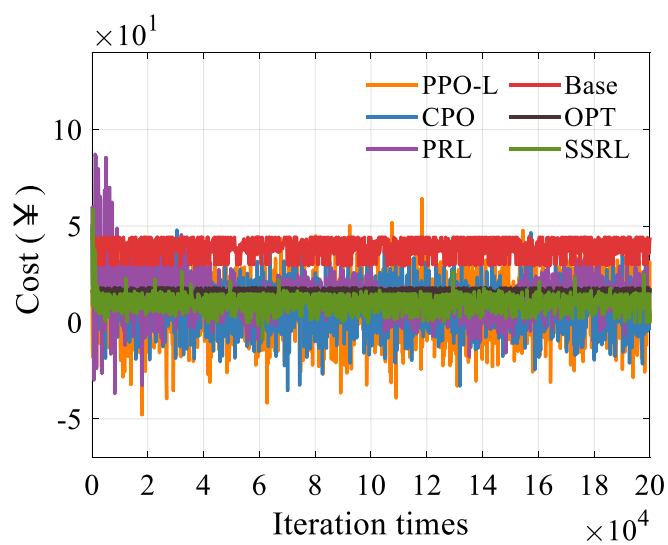

Fig. 7. Comparison of charging costs of the six methods.

Table 5
Comparison of the capabilities among the six methods.

Index	Abbr.	Method Description	Optimizing charging cost	Meeting EVs' SoC targets	Performing peak shaving	Ensuring operational security
1	Base	Uncontrolled charging (mostly used in the industry)		✔		
2	OPT	Optimization (using evolutionary algorithm)	✔	✔		
3	PRL	MAPPO approach based on penalty function method	✔	✔	✔	✔
4	PPO-L	PPO approach based on the Lagrangian method	✔	✔		
5	CPO	SRL's benchmark methods	✔	✔		
6	SSRL	The proposed approach in this paper	✔	✔	✔	✔

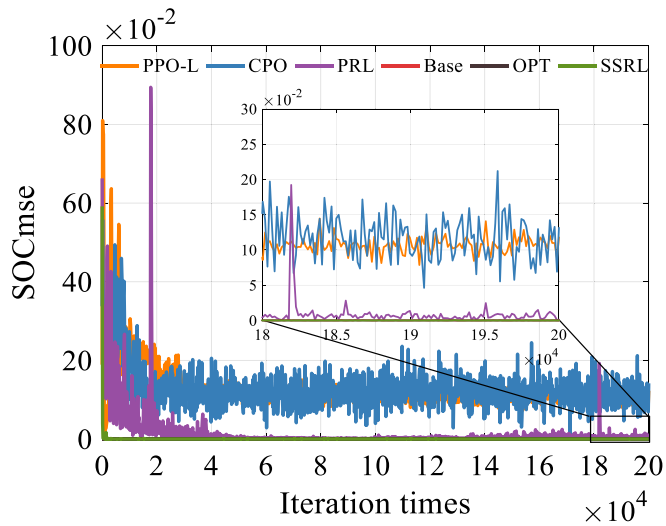

Fig. 8. Comparison of SoC deviation when the EV leaves.

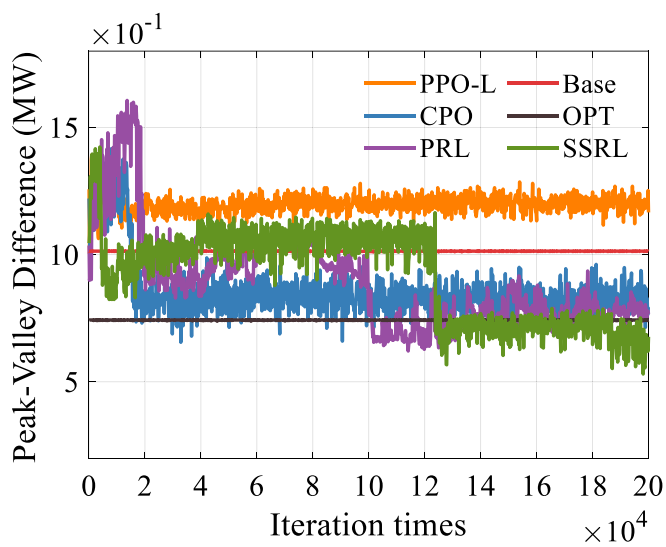

Fig. 9. Comparison of the maximum peak-valley load differences.

Fig. 9 compares the peak-valley difference (the difference between the maximum and the minimum load) in a complete charging cycle (showing by each point). It shows the results of "PPO-L(1.2 MW) > Base (1 MW) > CPO > OPT≈ PRL(0.75 MW) > the proposed approach". The reason is that PPO-L, Base, and CPO fail to incorporate the global objective of peak shaving, as indicated in Table 5. Although OPT does not directly optimize the global objective, it achieves peak shaving by responding to TOU prices. In contrast, PRL and the proposed approach outperform in finding optimal policies for peak shaving due to system-level reward. Fig. 10 compares the voltage deviation (9.1). PPO-L, OPT and CPO have voltage deviations due to partial observability. Compared with PRL, the voltage deviation of the proposed approach is lower and stable. This is owing to the protection of the global shield, which prevents bus voltage violations when searching for the optimal policy.

Figs. 9-10 demonstrate that at the system level, it is difficult for PRL to find a policy that satisfies the global security while optimizing the global objective. As a comparison, the proposed approach uses the global shield to guarantee the global security criterion such that the agent can achieve a better peak-shaving by finding a more optimal policy in the secure domain.

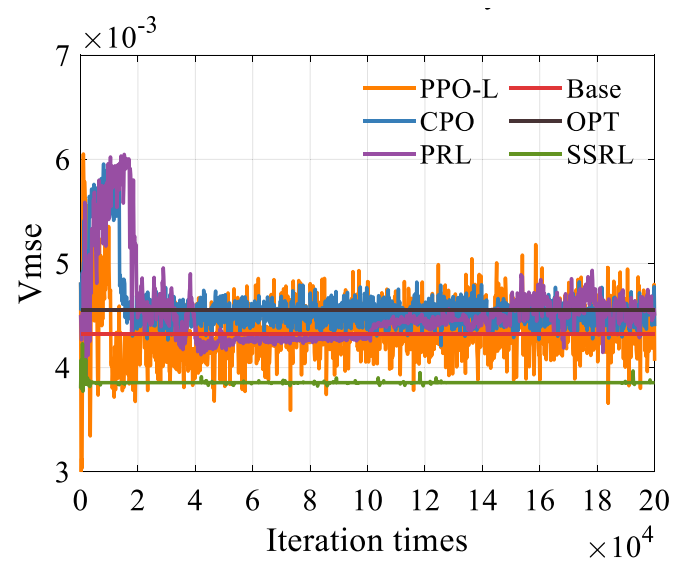

Fig. 10. Comparison of bus voltage deviations.

To verify the computational efficiency, Table 6 compares the average training time under all the six methods. The proposed SSRL achieves the fastest training, which is mainly due to the fact that it uses the shields to reduce the risky policy space. CPO takes a longer time due to the need to calculate conjugate gradients for approximating the fisher information matrix. Base and OPT are not shown as they don't need training.

4.4. Experimentation in the execution phase

The results in execution are shown in Figs. 11-13, where each horizontal axis represents the charging time period (i.e., from the earliest EV's arrival at 7:00 to the last EV's departure at 19:00). The same shields as in the training phase are used.

Fig. 11 shows the charging results of all the ten EV agents. Base, OPT and the proposed approach successfully meet the SoC targets for all EVs. In the other three methods, PRL fails for EV5, EV6, and EV7, PPO-L fails for EV4 and EV8, and CPO fails for EV4, EV5, and EV10, which is consistent with the results in Fig. 8 where these three methods cannot satisfy the EV's SoC target in the training. Moreover, due to the basic logic behind these methods, Base charges an EV at the maximum power immediately, and OPT charges it by taking full advantage of the TOU price variations (e.g., charges under the low price at 15:00). The charging patterns under PRL, PPO-L and CPO almost follow the TOU prices variation, but PRL performs better. CPO and PPO-L have errors in the approximation the constraint advantage function due to the sparse constraint signal. Hence, the agent cannot fully satisfy the EV-level constraints.

As a comparison, the merit of the proposed approach is clearly demonstrated. That is, the agent firstly discharges the EVs until reaching the lower SoC limit under the protection of the local shield, and when sensing the low TOU prices, it starts to charge the EV gradually (instead of a fast charging using the maximum power), which effectively avoids voltage deviation due to the protecting effect of the global shield.

Fig. 12 depicts the netload curves under the six methods. Only the proposed approach achieves peak shaving during 8:00–9:00 and 15:00–17:00. The five benchmark methods even exacerbate the peak-valley difference. In Base, the peak load occurs at 8:00 as most EVs

Table 6
Comparison of the average training time of all methods.

Methods	Base	OPT	PRL	PPO-L	CPO	SSRL
Average time (h)	/	/	17.35	16.42	22.92	7.82

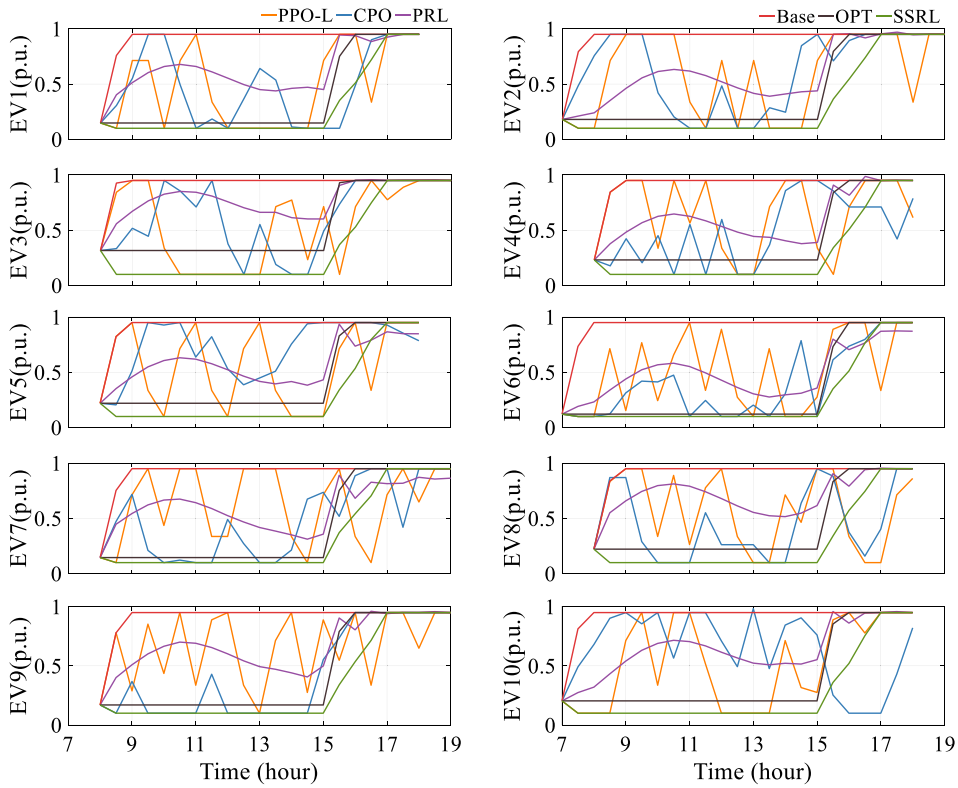

Fig. 11. Real-time charging process of all the ten EV agents.

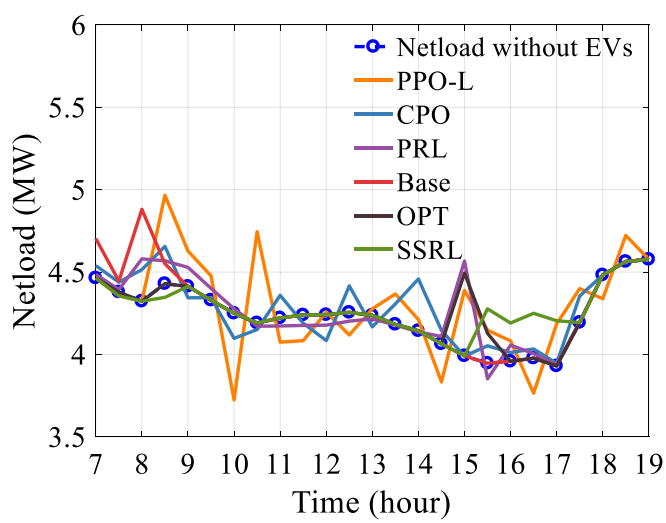

Fig. 12. Real-time variations of the system netloads.

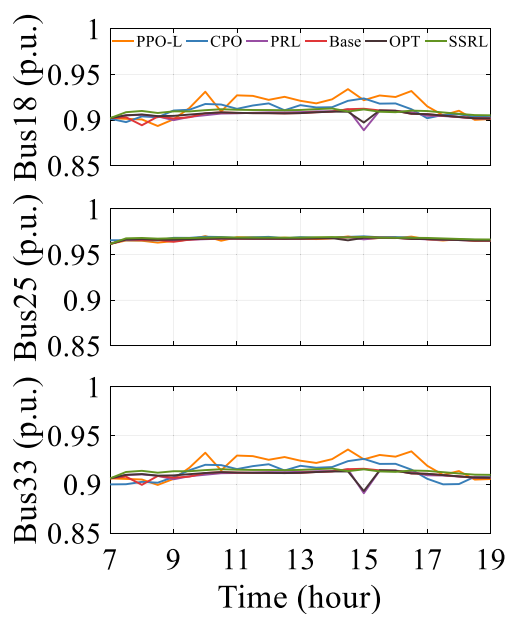

Fig. 13. Real-time voltage variations at the feeder-end buses.

arrive at this time. The other four benchmark methods perform most of the charging under low prices and thus cause the peak loads at 14:00–15:00.

Fig. 13 shows the voltages at the three buses located at the ends of the distribution feeders (see Fig. 6). All five benchmark methods result in undervoltage (at buses 18 and 33 at 8:00 and 15:00). As a comparison, the proposed approach always maintains these voltages within the acceptable ±10 %.

The results in Figs. 12-13 demonstrate that the benchmark methods except PRL have very poor performance regarding global objectives and global constraints due to their partial observability. PRL has a limited performance as it is difficult to reconcile the objectives and constraints at the global and local levels. The proposed approach uses the shields to prevent voltage violations (system-level constraints), and also achieves a better peak-shaving performance than the benchmark methods.

Table 7 lists the total charging cost for each EV in the entire charging cycle. The proposed approach has the lowest total cost, which is consistent with the result in Fig. 7. The following EVs are not shown in this Table 7 as they fail to meet their SoC target: EVs 5 ~ 7 in PRL, EV4 and EV8 in PPO-L, and EV4 ~ 5 and EV10 in CPO.

To verify the decision efficiency, Table 8 compares the execution times in a step and in an episode (a charging cycle) under all six

Table 7
Comparison of the total charging cost of all methods.

Total cost (¥)	Base	OPT	PRL	PPO-L	CPO	SSRL
EV1	39.36	15.06	25.64	13.44	17.61	13.97
EV2	37.77	14.46	20.85	17.96	9.37	12.63
EV3	31.1	11.90	21.44	8.89	11.62	6.98
EV4	32.28	13.50	16.59	/	/	10.52
EV5	35.83	13.71	/	32.94	/	10.98
EV6	40.65	15.56	/	30.63	30.79	15.07
EV7	39.46	15.10	/	16.58	20.43	14.06
EV8	35.73	13.67	20.33	/	36.72	10.90
EV9	38.29	14.65	22.92	41.00	11.33	13.07
EV10	36.6	14.00	22.97	20.93	/	11.64
Average	36.707	14.162	21.53	22.80	19.72	11.99

Table 8
Comparison of the average execution time of all methods.

Average time (ms)	Base	OPT	PRL	PPO-L	CPO	SSRL
One step	10.9	318.6	47.8	43.5	40.4	62.6
One episode	276	7646.4	1254.2	1189.4	1131.2	1546.3

methods. Base is the fastest. OPT takes the longest time mostly used by its evolutionary algorithm. The remaining methods including the proposed approach also take short execution times which are suitable for real-time decision making, whereas the proposed approach takes a slightly longer time due to the existence of the shield.

5. Conclusion

To address the PO-CMDP EV charging problem and to overcome the RL methods' most critical drawback—security risk, this paper proposes a rule-based shield embedded safe RL framework, which is implemented by a developed centralized-training and decentralized-execution MAPPO algorithm. The proposed framework not only uses rule-based shields to guarantee that the agent's actions strictly satisfy the local and global security criteria in the training and execution phases, but also helps the agent to find a more near-optimal policy.

The comparative study demonstrates that the proposed method achieves the best performance in terms of charging cost and peak shaving while strictly satisfying the EV SoC constraint and voltage constraint. Compared to CPO (the existing safe RL benchmark), the proposed approach not only reduces charging costs by 39 % and peak load by 19 %, but also maintains reasonable violation levels of both local and global security constraints in both training and execution phases. This shows that the proposed approach is a promising way to realize efficient and safe EV charging scheduling.

Future research can focus on the following topics. 1) Model-based RL: As system scale expands, acquiring accurate physical rules for constructing shields may be more challenging. This can be addressed by Model-based RL which constructs a security domain by using the interaction data between agents and the environment and reduces the reliance on prior knowledge. 2) Scaling-up safe RL: Although the rule-based shields safe RL is a distributed and scalable approach, it is necessary to study the representation of large state and action spaces, especially in the cases of large-scale distributed resources in a grid. 3) Knowledge transferring: One can leverage the knowledge (i.e., policy or value function) learned in EV dispatch in the problems of scheduling other distributed resources for accelerating the convergence in the agents' training.

CRediT authorship contribution statement

Yuxiang Guan: Conceptualization, Methodology, Software, Validation, Visualization, Writing – original draft. **Jin Zhang:** Data curation, Investigation, Validation. **Wenhao Ma:** Data curation, Investigation, Validation. **Liang Che:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix

Theorem 2. *Given the case mentioned in Theorem 1, the variance of estimating the expected cumulative reward in the proposed method is much smaller than that in the Lagrangian method, i.e.,* $\mathrm{Var}[J_{shield}(\pi_\theta)] < \mathrm{Var}[J_{Lag}(\pi_\theta)]$.

Proof: The proof is this Theorem is based on the following relationships:

$$\mathrm{Var}[J_{shield}(\pi_\theta)] = \mathrm{Var}\left[\sum_{t=0}^{\infty}\gamma^t r_t^{shield}\right]$$

$$= \frac{1}{(1-\gamma)^2}\bullet\left[\begin{array}{l}\left(r_t - E_{\tau\ \pi}\left[r_t^{shield}\right]\right)^2\bullet\left(1-\mathscr{P}_{Lag}\right)\\ +\left(r_t^{shield} - E_{\tau\ \pi}\left[r_t^{shield}\right]\right)^2\bullet\mathscr{P}_{Lag}\end{array}\right]$$

$$\le \frac{1}{(1-\gamma)^2}\left((2)^2\left(1-\mathscr{P}_{Lag}\right)+(2)^2\mathscr{P}_{Lag}\right) = \frac{4}{(1-\gamma)^2}$$

$$< \frac{1}{(1-\gamma)^2}\left(\mathscr{P}_{Lag}\left(\xi c_{Lag}\right)^2 - \left(\mathscr{P}_{Lag}\xi c_{Lag}\right)^2\right) \le \mathrm{Var}\left[J_{Lag}(\pi_\theta)\right] \tag{30.1}$$

The first inequality sign is proved as follows. Since the reward provided by the environment and the safety penalty imposed by the shield are both in the range of $[-1, 1]$, the maximum deviation of R or R^{shield} is $1-(-1) = 2$. So,

$$\begin{cases} r_t - E_{\tau\ \pi}\left[r_t^{shield}\right] \le 2 \\ r_t^{shield} - E_{\tau\ \pi}\left[r_t^{shield}\right] \le 2 \end{cases} \tag{30.2}$$

Applying (30.2) to the 3rd line of (30.1) proves the first inequality sign. The second inequality sign in (30.1) is proved as follows. Due to the PO-CMDP's large policy space, among all the trajectories, the probability of simultaneously charging/discharging all EVs at their maximum powers and causing security issue is very low, i.e., $\mathscr{P}_{Lag}$ is very small. For brevity, it can be represented by $\mathscr{P}_{Lag} \le 1/\mathscr{M}$, where $\mathscr{M}$ is a very large number. And based on Lemma 1 ($\xi c_{Lag} \gg 1/\mathscr{P}_{Lag}$), one has $\xi c_{Lag} \gg \mathscr{M}$. So,

$$\mathscr{P}_{Lag}\left(\xi c_{Lag}\right)^2 - \left(\mathscr{P}_{Lag}\xi c_{Lag}\right)^2 = \mathscr{P}_{Lag}\left(\xi c_{Lag}\right)^2\left(1 - \mathscr{P}_{Lag}\right)$$

$$\gg \frac{1}{\mathscr{P}_{Lag}}\left(1 - \mathscr{P}_{Lag}\right) \gg \mathscr{M}\left(1 - \frac{1}{\mathscr{M}}\right) = \mathscr{M} - 1 \tag{30.3}$$

As $\mathscr{M}$ is a very large number, it can be ensured that the left-hand side of (30.3) is larger than 4. This proves the second inequality sign in (30.1). Finally, the third inequality sign in (30.1) has been proved in Theorem 1.

References

[1] International Energy Agency (IEA), Global EV Outlook 2023, Paris, France, April 2023. [Online]. Available: https://www.iea.org/reports/global-ev-outlook-2023.

[2] Moschella M, Murad MAA, Crisostomi E, et al. Decentralized charging of plug-in electric vehicles and impact on transmission system dynamics. IEEE Trans Smart Grid 2020;12(2):1772–81. https://doi.org/10.1109/TSG.2020.3034528.

[3] Cao Y, Huang L, Li Y, et al. Optimal scheduling of electric vehicles aggregator under market price uncertainty using robust optimization technique. Int J Electr Power Energy Syst 2020;117:105628. https://doi.org/10.1016/j.ijepes.2019.105628.

[4] Naderi E, Pourakbari-Kasmaei M, Cerna FV, et al. A novel hybrid self-adaptive heuristic algorithm to handle single-and multi-objective optimal power flow problems. Int J Electr Power Energy Syst 2021;125:106492. https://doi.org/10.1016/j.ijepes.2020.106492.

[5] Cao D, Hu W, Zhao J, et al. Reinforcement learning and its applications in modern power and energy systems: A review. J Modern Power Syst Clean Energy, 2020; 8 (6): 1029-1042. 10.35833/mpce.2020.000552.

[6] Dabbaghjamanesh M, Moeini A, Kavousi-Fard A. Reinforcement learning-based load forecasting of electric vehicle charging station using Q-learning technique. IEEE Trans Ind Inf 2020;17(6):4229–37. https://doi.org/10.1109/TII.2020.2990397.

[7] Liang Y, Ding Z, Ding T, et al. Mobility-aware charging scheduling for shared on-demand electric vehicle fleet using deep reinforcement learning. IEEE Trans Smart Grid 2020;12(2):1380–93. https://doi.org/10.1109/TSG.2020.3025082.

[8] Liang Y, Ding Z, Zhao T, et al. Real-time operation management for battery swapping-charging system via multi-agent deep reinforcement learning. IEEE Trans Smart Grid 2023;14(1):559–71. https://doi.org/10.1109/TSG.2022.3186931.

[9] Ye Z, Gao Y, Yu N. Learning to operate an electric vehicle charging station considering vehicle-grid integration. IEEE Trans Smart Grid 2022;13(4):3038–48. https://doi.org/10.1109/TSG.2022.3165479.

[10] Shin MJ, Choi DH, Kim J. Cooperative management for PV/ESS-enabled electric vehicle charging stations: A multiagent deep reinforcement learning approach. IEEE Trans Ind Inf 2019;16(5):3493–503. https://doi.org/10.1109/TII.2019.2944183.

[11] Li H, Li G, Lie TT, et al. Constrained large-scale real-time EV scheduling based on recurrent deep reinforcement learning. Int J Electr Power Energy Syst 2023;144: 108603. https://doi.org/10.1016/j.ijepes.2022.108603.

[12] Li S, et al. EV charging strategy considering transformer lifetime via evolutionary curriculum learning-based multiagent deep reinforcement learning. IEEE Trans Smart Grid 2022;13(4):2774–87. https://doi.org/10.1109/TSG.2022.3167021.

[13] Yan L, Chen X, Chen Y, et al. A cooperative charging control strategy for electric vehicles based on multiagent deep reinforcement learning. IEEE Trans Ind Inf 2022;18(12):8765–75. https://doi.org/10.1109/TII.2022.3152218.

[14] Xing Q, Xu Y, Chen Z, et al. A graph reinforcement learning-based decision-making platform for real-time charging navigation of urban electric vehicles. IEEE Trans Ind Inf 2022;19(3):3284–95. https://doi.org/10.1109/TII.2022.3210264.

[15] Xing Q, Xu Y, Chen Z. A bilevel graph reinforcement learning method for electric vehicle fleet charging guidance. IEEE Trans Smart Grid 2023;14(4):3309–12. https://doi.org/10.1109/TSG.2023.3240580.

[16] Li S, Hu W, Cao D, et al. A multiagent deep reinforcement learning based approach for the optimization of transformer life using coordinated electric vehicles. IEEE Trans Ind Inf 2022;18(11):7639–52. https://doi.org/10.1109/TII.2021.3139650.

[17] Dorokhova M, Martinson Y, Ballif C, et al. Deep reinforcement learning control of electric vehicle charging in the presence of photovoltaic generation. Appl Energy 2021;301:117504. https://doi.org/10.1016/j.apenergy.2021.117504.

[18] Da Silva FL, Nishida CEH, Roijers DM, et al. Coordination of electric vehicle charging through multiagent reinforcement learning. IEEE Trans Smart Grid 2019; 11(3):2347–56. https://doi.org/10.1109/TSG.2019.2952331.

[19] Wang W, Yu N, Gao Y, et al. Safe off-policy deep reinforcement learning algorithm for volt-var control in power distribution systems. IEEE Trans Smart Grid 2019;11 (4):3008–18. https://doi.org/10.1109/TSG.2019.2962625.

[20] Ye Y, Wang H, Chen P, et al. Safe deep reinforcement learning for microgrid energy management in distribution networks with leveraged spatial-temporal perception. IEEE Trans Smart Grid 2023;14(5):3759–75. https://doi.org/10.1109/TSG.2023.3243170.

[21] Shi X, Xu Y, Chen G, et al. An augmented lagrangian-based safe reinforcement learning algorithm for carbon-oriented optimal scheduling of ev aggregators. IEEE Trans Smart Grid 2023. https://doi.org/10.1109/TSG.2023.3289211.

[22] Yi H, Park E, Kim S. Multi-agent deep reinforcement learning for autonomous driving. KIISE Transactions on Computing Practices 2018;24(12):670–4.

[23] J. Achiam, D. Held, A. Tamar, and P. Abbeel. Constrained Policy Optimization. 34th International Conference on Machine Learning. ICML 2017, 1:30-47, 2017. http://arxiv.org/abs/1705.10528.

[24] Yang TY, Rosca J, Narasimhan K, et al. Projection-based constrained policy optimization. arXiv preprint arXiv:2010.03152, 2020. https://arxiv. org/abs/2010.03152.

[25] Li H, Wan Z, He H. Constrained EV charging scheduling based on safe deep reinforcement learning. IEEE Trans Smart Grid 2019;11(3):2427–39. https://doi.org/10.1109/TSG.2019.2955437.

[26] Li H, He H. Learning to operate distribution networks with safe deep reinforcement learning. IEEE Trans Smart Grid 2022;13(3):1860–72. https://doi.org/10.1109/TSG.2022.3142961.

[27] Chen P, Liu S, Wang X, et al. Physics-Shielded multi-agent deep reinforcement learning for safe active voltage control with photovoltaic/battery energy storage systems. IEEE Trans Smart Grid 2022;11(4):2656–67. https://doi.org/10.1109/TSG.2022.3228636.

[28] Zhao H, Zhao J, Qiu J, et al. Cooperative wind farm control with deep reinforcement learning and knowledge-assisted learning. IEEE Trans Ind Inf 2020; 16(11):6912–21. https://doi.org/10.1109/TII.2020.2974037.

[29] Tao Y, Qiu J, Lai S, et al. A human-machine reinforcement learning method for cooperative energy management. IEEE Trans Ind Inf 2021;18(5):2974–85. https://doi.org/10.1109/TII.2021.3105115.

[30] Hernandez-Leal P, Kartal B, Taylor ME. A survey and critique of multiagent deep reinforcement learning. Auton Agent Multi-Agent Syst 2019;33(6):750–97. https://doi.org/10.1007/s10458-019-09421-1.

[31] Chandra Mouli G, Kefayati M, Baldick R, et al. Integrated PV Charging of EV Fleet Based on Energy Prices, V2G, and Offer of Reserves. IEEE Trans Smart Grid 2019; 10(2):1313–25. https://doi.org/10.1109/TSG.2017.2763683.

[32] Fretzen U, Ansarin M, Brandt T. Temporal city-scale matching of solar photovoltaic generation and electric vehicle charging. Appl Energy 2021;282:116160. https://doi.org/10.1016/j.apenergy.2020.116160.

[33] Zhang J, Guan Y, Che L, et al. EV Charging Command Fast Allocation Approach Based on Deep Reinforcement Learning with Safety Modules. IEEE Trans Smart Grid 2024;15(1):757–69. https://doi.org/10.1109/TSG.2023.3281782.

[34] Xu T, Li Z, Yu Y. Error bounds of imitating policies and environments for reinforcement learning. IEEE Trans Pattern Anal Mach Intell 2021;44(10): 6968–80. https://doi.org/10.1109/TPAMI.2021.3096966.

[35] Yang L, Ji J, Dai J, et al. Cup: A conservative update policy algorithm for safe reinforcement learning. arXiv preprint arXiv:2202.07565, 2022. https://arxiv.org/abs/2202.07565.

[36] Dalal G, Dvijotham K, Vecerik M, et al. Safe exploration in continuous action spaces. arXiv preprint arXiv:1801.08757, 2018. https://arxiv. org/abs/1801.08757.

[37] J. Schulman, P. Moritz, S. Levine, M. I. Jordan, and P. Abbeel. High-dimensional continuous control using generalized advantage estimation. 4th International Conference on Learning Representations. ICLR 2016, San Juan, Puerto rico, pp:1-14, 2016. https://arxiv.org/pdf/1506.02438.

[38] Wang S, Liu Q, Ji X. A fast sensitivity method for determining line loss and node voltages in active distribution network. IEEE Trans Power Syst 2017;33(1): 1148–50. https://doi.org/10.1109/TPWRS.2017.2735898.

[39] Wang J, Xu W, Gu Y, et al. Multi-agent reinforcement learning for active voltage control on power distribution networks. Advances in Neural Information Processing Systems, 2021, 34: 3271-3284. 10.48550/arXiv.2110.14300.

[40] Yu C, Velu A, Vinitsky E, et al. The surprising effectiveness of ppo in cooperative multi-agent games. Advances in Neural Information Processing Systems, 2022, 35: 24611-24624. https://arxiv.org/abs/2103.01955.

[41] Shi J, Gao Y, Wang W, et al. Operating electric vehicle fleet for ride-hailing services with reinforcement learning. IEEE Trans Intell Transp Syst 2019;21(11):4822–34. https://doi.org/10.1109/TITS.2019.2947408.
[42] California ISO, Open Access Same-Time Inf. Syst., Folsom, CA, USA. [Online]. Available: http://oasis.caiso.com/mrioasis/logon.do.
[43] Santos A, McGuckin N, Nakamoto HY, et al. Summary of travel trends: 2009 national household travel survey. United States: Federal Highway Administration; 2011.